

Flowserve PVXM-3 Centrifugal Pump

Operation, Maintenance & Troubleshooting Manual

Applicable Units: P-01, P-02, P-07, P-10 (CDU Service)

1. Pump Specifications

Parameter	Value
Model	PVXM-3
Type	Single-stage, end-suction, centrifugal
Service fluid	Crude oil / heavy hydrocarbons
Design flow	120 m ³ /hr
Design head	85 m
Rated power	37 kW
Rated speed	2,950 RPM
Specific speed (Ns)	1,850 (metric)
Impeller diameter	285 mm
Impeller type	Closed, 5-vane
Casing material	Carbon steel ASTM A216 WCB
Impeller material	316 stainless steel
Shaft material	17-4 PH stainless steel
Bearing type	SKF 6312 (drive end), SKF 6308 (non-drive end)
Seal type	Mechanical double seal, API Plan 53B
Seal flush fluid	Barrier fluid (glycol/water 50:50)
NPSH required	3.2 m at design flow
Maximum allowable temperature	180°C
Maximum allowable pressure	16 bar

2. Normal Operating Envelope

2.1 Flow & Head Envelope

- Minimum continuous flow: 48 m³/hr (40% of BEP)
- Best Efficiency Point (BEP): 120 m³/hr at 85 m head
- Maximum continuous flow: 138 m³/hr (115% of BEP)
- Operating outside this range causes increased vibration, bearing loads, and seal wear

2.2 Bearing Temperature Limits

Condition	Bearing Temp (°C)	Action
Normal operation	45–65°C	No action required
Elevated — monitor	65–80°C	Increase monitoring frequency; check lubrication
Warning threshold	80–85°C	Inspect bearing and lubrication within 4 hours
Alarm threshold	>85°C	Reduce load or shut down; inspect immediately
Failure threshold	>95°C	Immediate shutdown — bearing failure imminent

Rate-of-rise alert: If bearing temperature rises >2°C/hr sustained over 4 hours, treat as warning regardless of absolute temperature.

2.3 Vibration Limits (ISO 10816-3, Group 2, >15 kW)

Zone	Vibration (mm/s RMS)	Condition	Action
A	0 – 2.3	New or recently overhauled	Normal
B	2.3 – 4.5	Acceptable for long-term operation	Monitor
C	4.5 – 7.1	Unsatisfactory — investigate	Plan maintenance within 2 weeks
D	> 7.1	Dangerous — damage likely	Shut down immediately

Measurement points: - Bearing housing, radial direction (X and Y axes) - Bearing housing, axial direction - Measure at BEP flow conditions for baseline

2.4 Motor Current Limits

- Normal operating current: 58–65 A (at rated conditions)
 - Overload warning: >72 A sustained >5 minutes
 - Overload trip: >80 A (motor protection relay setting)
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3. Bearing Maintenance

3.1 Lubrication Specifications

- Lubricant type: Mobil SHC 526 (synthetic, ISO VG 100)
- Grease alternative: Shell Gadus S2 V220 (NLGI Grade 2)
- Relubrication interval: every 2,000 operating hours
- Grease quantity per bearing: 15–20 g (do not over-grease — causes overheating)
- Operating temperature range for lubricant: –30°C to +130°C

3.2 Bearing Replacement Criteria

Replace bearings when any of the following occur: - Bearing temperature consistently >80°C despite fresh lubrication - Vibration in Zone C or D - Audible noise: grinding, squealing, or irregular knocking - Visible pitting, spalling, or discoloration on races or rolling elements - Radial clearance exceeds 0.08 mm (measured with feeler gauge) - After any bearing temperature excursion >95°C

3.3 Bearing Replacement Procedure

1. Isolate pump from process (close suction and discharge valves)
2. Drain seal barrier fluid; depressurize seal system
3. Remove coupling guard and disconnect coupling
4. Remove bearing housing cover bolts (torque: 45 N·m)
5. Extract shaft assembly using bearing puller — do NOT use heat on SKF 6312
6. Press new bearing onto shaft using bearing heater (max 110°C) or press
7. Verify bearing seating: zero axial play, smooth rotation by hand
8. Repack with fresh grease (15 g per bearing)
9. Reassemble; align coupling to ≤ 0.05 mm TIR
10. Run-in: operate at 50% load for 2 hours; check temperature and vibration

3.4 Bearing Life Estimation (L10 Life)

Using SKF bearing life formula:

$$L_{10} \text{ (hours)} = (C/P)^3 \times (10^6 / (60 \times n))$$

Where: - C = dynamic load rating (SKF 6312: 81.9 kN) - P = equivalent dynamic bearing load (kN) - n = shaft speed (RPM)

At design conditions ($P \approx 8.5$ kN, $n = 2,950$ RPM): - $L_{10} \approx 42,000$ hours (~5 years continuous operation) - Actual service life reduced by: misalignment, contamination, overtemperature

4. Mechanical Seal Maintenance

4.1 Seal System — API Plan 53B

- Barrier fluid: Glycol/water 50:50 (inhibited)
- Barrier fluid pressure: 2–3 bar above process pressure
- Barrier fluid temperature: maintain $<60^{\circ}\text{C}$ (cooling water required if $>50^{\circ}\text{C}$ ambient)
- Reservoir volume: 2.5 liters
- Refill interval: check monthly; refill if level drops $>10\%$

4.2 Seal Failure Indicators

- Seal leak detected (proximity sensor): immediate action required
- Barrier fluid consumption >0.5 L/week: inspect seal faces
- Barrier fluid pressure drop >0.5 bar/day: seal face or O-ring failure
- Visible leakage at seal housing: shutdown required

4.3 Seal Replacement Procedure

1. Shut down pump; isolate from process
2. Depressurize and drain barrier fluid system
3. Remove coupling and bearing housing
4. Slide seal cartridge off shaft (cartridge design — no special tools required)
5. Inspect seal faces: replace if scratched, chipped, or worn >0.5 mm
6. Install new cartridge seal; verify O-ring seating
7. Refill barrier fluid; pressurize to 2 bar; check for leaks
8. Typical seal replacement time: 4–6 hours

4.4 Seal Life Expectancy

- Normal service: 18–24 months
 - Reduced life factors: dry running (even momentary), process fluid contamination of barrier fluid, temperature $>150^{\circ}\text{C}$, shaft misalignment >0.1 mm
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5. Impeller Inspection & Replacement

5.1 Impeller Wear Indicators

- Differential pressure declining $>10\%$ from baseline at same flow: impeller wear
- Flow rate declining $>15\%$ at same speed and head: impeller damage

- Visible erosion on vane leading edges (inspect during overhaul)
- Cavitation damage: pitting on suction side of vanes

5.2 Wear Ring Clearance

- New clearance (impeller/casing wear ring): 0.35–0.45 mm
- Maximum allowable clearance: 0.80 mm
- If clearance >0.80 mm: replace wear rings (do not replace impeller unless damaged)

5.3 Impeller Replacement

1. Remove pump casing (8 × M20 bolts, torque 120 N·m)
2. Lock shaft; remove impeller nut (left-hand thread — turn clockwise to remove)
3. Pull impeller; inspect for erosion, cracks, or imbalance
4. Install new impeller; torque nut to 85 N·m
5. Check axial clearance between impeller and casing: 0.5–1.0 mm
6. Reassemble; perform vibration check at startup

6. Troubleshooting Guide

Symptom	Probable Cause	Corrective Action
Bearing temp >80°C	Insufficient lubrication	Re-grease; check interval
Bearing temp >80°C	Misalignment	Check and correct coupling alignment
Bearing temp >80°C	Bearing failure	Replace bearing
Vibration Zone C/D	Impeller imbalance or damage	Inspect impeller; balance or replace
Vibration Zone C/D	Misalignment	Realign coupling
Vibration Zone C/D	Cavitation	Check NPSH; reduce flow or increase suction pressure
Vibration axial high	Misalignment (angular)	Check angular alignment
Flow rate declining	Impeller wear	Check wear ring clearance; replace if needed
Seal leak	Seal face damage	Replace mechanical seal
Motor overcurrent	Impeller rubbing casing	Check axial clearance
Noise — cavitation	NPSH insufficient	Increase suction pressure; reduce flow
Noise — bearing	Bearing failure	Replace bearing immediately